

BIRZEIT UNIVERSITY

Faculty of Information Technology
Computer Science Department

Web Applications and Technologies - COMP334 Midterm Exam

Date: Sunday, 04/05/2014

Choose Your Instructor: ☐ Dr. Yousef A Hassouneh

Time: 80 minutes

☒ Mr. Ibrahim M Srahin

Student Number:	Sara Abu Hijleh
Student Name:	1110348

Instructions:

- ✓ This is a closed book exam. Answer all of the four questions, no optional questions.
- ✓ Answer the questions clearly in the space provided. Use the back of the pages for your rough work
- ✓ Students' questions are allowed only in the first 10 minutes after distribution the exam papers.
- ✓ Leaving the exam not before half the exam time being passed.

Question #	1	2	3	4	5	Total of 100
Grade	16	10	18	20	15	79

Question # 1 (20 points) :

A. Describe the components of a typical three-tier web application.

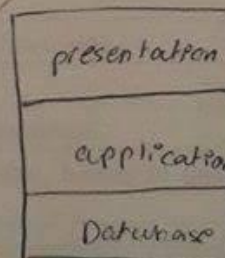
client (browser) :

- 1- receive request
- 2- received event
- 3- form submission

resource :
implement resource on server (store)

server :

- 1- implement access to data
- 2- implement security, efficiency
- 3- make a response
- 4- receive submission form



Question 4) (20 points)

2. Write MATLAB codes that process the following case of a histogram

a

Case
1. Histogram of a variable
2. Histogram of a variable
3. Histogram of a variable
4. Histogram of a variable
5. Histogram of a variable
6. Histogram of a variable
7. Histogram of a variable
8. Histogram of a variable
9. Histogram of a variable
10. Histogram of a variable

Case 1

1. Histogram of a variable

2. Histogram of a variable

3. Histogram of a variable

4. Histogram of a variable

5. Histogram of a variable

6. Histogram of a variable

Case 2

1. Histogram of a variable

2. Histogram of a variable

3. Histogram of a variable

4. Histogram of a variable

5. Histogram of a variable

6. Histogram of a variable

7. Histogram of a variable

8. Histogram of a variable

9. Histogram of a variable

10. Histogram of a variable

Case 3

1. Histogram of a variable

2. Histogram of a variable

3. Histogram of a variable

Case 4

1. Histogram of a variable

2. Histogram of a variable

3. Histogram of a variable

Question # 4 (25 points) :

Write a suitable CSS rule for each of the following styles.

- A. All Heading at level 2 through 5 should be in center alignment.

```
h2, h3, h4, h5 { align = center;
                }
```

- B. Each paragraph should be preceded by a 3 cm indentation and 2 points blue border .

```
P { padding-left = 3cm; "outside"
  border: 2;
  border-color: blue;
}
```

- C. Entities in class "first" should have a black color background and a white font.

```
.first { background-color = black;
        font-color = white; }
```

Question # 5 (25 points) :

Assume you have the following table called cars in the database:

CNo	CarModel	RaceYear	Driver	place
11	BMW	2001	Yousef	2
22	Skoda	2001	Ibrahim	1
33	VW	2001	Iyad	3
44	BMW	2002	Mustafa	1
55	VW	2002	Ali	2

Write a PHP scripts that allows the user to select a race year from a Combo Box then after submission the script should display a list of cars that have participated in the selected year.

Notes:

- The list of cars should have (Car Model, Driver and Place) and it should be in ascending order according to the place. ✓
- You have to use session to store the year.
- Assume the database name is race, username: httpd, password: test and server: localhost ✓

<?php>

\$host = 'localhost';

\$username = 'httpd';

\$password = 'test';

\$dbname = 'race';

\$link = sql_connect(localhost, httpd, test);

while (\$row = sql_fetch(\$link)) {

<table>

echo "<tr>";

echo "<td>\$carModel</td>";

echo "<td> Race year</td>";

Sort
echo "<td> place</td>";

echo "</tr>";

echo "<tr>";

echo "<td>". \$row[CarModel]. "</td>";

echo "<td>". \$row[Race year]. "</td>";

echo "<td>". ~~\$place~~ \$row[place]. "</td>";

echo "</tr>";

</table>



after
the
to next
show page
all the
car
from
database

B. Write XHTML codes that generate the following table:

Then apply the following CSS rules (in the header section of the page):

- Caption should be centered with Times New Roman font. ✓
- Horizontal table header should be centered, bold and red.
- Vertical table header text color should be yellow with green background.
- Table data cells should be italics and centered.

| Tasks Schedule | | | | |
|----------------|------|------|----------|------|
| | Day1 | Day2 | Day3 | Day4 |
| Task1 | T11 | T12 | Revision | |
| Task2 | T2 | | | |
| Task3 | T31 | T32 | | |

~~<link rel="stylesheet" type="text/css" style="font-family: 'Times New Roman'; align: center; }~~

~~#horiz { align: center; color: red; text-align: bold; }~~

~~#V { color: yellow; Background: 'green'; }~~

~~td { font-style: italic; align: center; }~~

<html>

<body>

<p> Tasks schedule </p>

<table>

<tr id="horiz">

<td> </td>

<th> Day1 </th>

<th> Day2 </th>

<th> Day3 </th>

~~colspan="2" <th> Day 4 </th>~~

</tr>

<tr>

<th id="V"> ~~Task 1~~ </th>

<td> T11 </td>

<td colspan="2"> T12 </td>

~~rowspan="3" <td> Revision </td>~~

</tr>

```

        </tr>
<th id='v'>Task 2 Task 2 </th>
        <td colspan='3'>T2 </td>
    </tr>
    <tr>
<th id='v'>Task 3 Task 3 </th>
        <td>T31 </td>
        <td>T32 </td>
        <td>T33 </td>
    </tr>
</body>
</html>

```

C. Entities in class "first" should have a black color background and a white font.

• first { background-color = black;
font-color = white; }

D. A hyperlink that is active should appear on orange and larger by 1.5 em.

a { size = 1.5em;
background-color = orange; }

E. All paragraph that have the "later" class should be hidden.

p.later { status = hidden; }

Question # 4 (25 points) :

Write a suitable CSS rule for each of the following styles.

- A. All Heading at level 2 through 5 should be in center alignment.

```
h2, h3, h4, h5 { align = center;
}
```

- B. Each paragraph should be preceded by a 3 cm indentation and 2 points blue border .

```
p { float = 3 cm; "all side"
  border = 2;
  border-color = blue;
}
```

- C. Entities in class "first" should have a black color background and a white font.

```
.first { background-color = black;
        font-color = white; }
```

- D. A hyperlink that is active should appear on orange and larger by 1.5 em.

```
a { size = 1.5 em;
    font-color = orange; }
```

- E. All paragraph that have the "later" class should be hidden.

```
p.later { status = hidden; }
```


g. Explain what the various mechanisms and how they're different types of cookies

The mechanism for cookies is to store the data of the client on the server. It's like the ~~server~~ client has not or single thing that, but it's not what you want to be stored on the client. It's not the server's data, but it's the client's data. The server is the one that stores the data, but the client is the one that sends the data to the server.

The difference between GET and POST methods

GET is for the client to send the data to the server. POST is for the client to send the data to the server.

c. What is the difference between GET and POST methods?

1. GET is for the client to send the data to the server. POST is for the client to send the data to the server.

2. POST is for the client to send the data to the server. GET is for the client to send the data to the server.

d. Identify and describe the different components parts of the following URL:

http://www.example.com:8080/path/to/resource

1. http: The protocol that will be used to access the resource.

2. www: The name of the site or the resource - help to identify the site.

3. 8080: The port of the page that it's required. It's going to be a port.

Question 2 (10 points)

Determine whether each of the following statements are TRUE or FALSE.

| Q | Statement |
|-------------------------------------|---|
| ☒ | 1. <u>www</u> field is larger than <u>http</u> . |
| ☒ | 2. <u>www</u> field is larger than <u>http</u> . |
| ☒ | 3. <u>www</u> field is much more used than <u>http</u> , leading to more and more resources in a network. |
| ☒ | 4. <u>www</u> and <u>http</u> can be used together to identify a web page on a web server. |
| ☒ | 5. The <u>www</u> field should be placed in the beginning of the <u>http</u> field. |
| ☒ | 6. <u>www</u> field is much more used. |